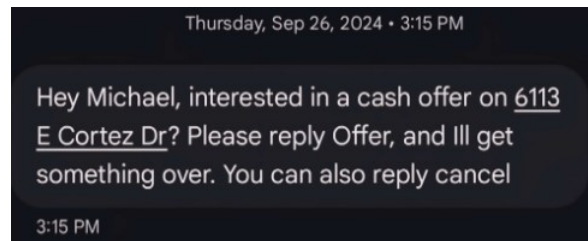


Detecting Spam Messages Using Machine Learning



Have you ever received a spam message over text like the one shown above? It seems like no matter how much you try to protect your phone number and personal information, they always reach you, sometimes many times a day. Besides being a nuisance, these messages can be harmful and dangerous, especially to less technologically literate populations. Spam messages can spread misinformation, especially politically related, and contain dangerous links to malware. Many spam messages also promote scams, where they intentionally mislead receivers in attempts to steal information or money, often preying on miscommunications and emotional scenarios.

Over time, these scam messages have evolved and have become more difficult to identify, often attempting to evoke feelings of urgency or emotion to make the recipient act fast [1]. This makes it important to identify the structural features of these predatory texts and irritating spam messages. Recent advances in machine learning have enabled models to classify texts based on patterns in wording structure and sentiment. Prior research shows that spam and phishing messages contain distinctive linguistic features such as excessive urgency, requests for personal information, and uncommon formatting [2]. By applying machine learning techniques to spam and “ham” messages it is possible to build systems that flags potentially dangerous messages in real time.

Your task, should you choose to accept, is to train a Naïve Bayes Classifier to categorize English SMS text messages as spam or ham with a high (~90%) success rate. You will begin with data cleaning, removing messages with non-readable characters and fixing decoding errors. Next, you will perform EDA focused on finding key distinguishing features between the message types, such as specific words or punctuation. Next, you will split the data and train the Count Vectorizer unigrams and bigrams on the training data and any additional features you choose to include, ensuring stop words are removed. Lastly, you will test your model on the testing data and any other personal messages you may wish to pass through. The model you make, focusing on efficiency and size, can help protect users from annoying and malicious messages.

References:

- [1] “How to spot a scam email, text message or call,” NCSC, <https://www.ncsc.gov.uk/collection/phishing-scams/spot-scams> (accessed Jan. 28, 2026).
- [2] Tiago A. Almeida, José María G. Hidalgo, and Akebo Yamakami. 2011. Contributions to the study of SMS spam filtering: new collection and results. In Proceedings of the 11th ACM symposium on Document engineering (DocEng '11). Association for Computing Machinery, New York, NY, USA, 259–262. <https://doi.org/10.1145/2034691.2034742>